جامعة الملك سعود
King Saud University
كلية علوم الحاسب والمعلومات
قسم تقنية المعلومات



REPLICA

Sprint-0 Report

Prepared by

Reem Almaskati	445201439
Sadeem Al Sultan	445201370
Sondos Aldughaim	445202340
Danah AlDhowaihy	445202207

Supervised by

Dr. Sharefah Alghamdi

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1 Introduction

Alzheimer's disease does not begin on the day a patient receives a diagnosis. Changes in the brain associated with Alzheimer's disease (AD) can begin a decade or more before noticeable problems with memory and thinking appear [1]. This long and gradual disease process makes Alzheimer's particularly difficult to monitor because a patient may be changing biologically and cognitively long before those changes become obvious in daily life. Alzheimer's disease is the most common cause of dementia, contributing to approximately 60–70% of dementia cases. In 2021, an estimated 57 million people were living with dementia worldwide, highlighting the growing impact of cognitive disorders on patients, families, healthcare systems, and communities [2].

Understanding how an individual patient is changing over time is therefore an important part of Alzheimer's research and care. Cognitive status is not the same for every patient. Some individuals remain cognitively unimpaired, while others develop Mild Cognitive Impairment (MCI), a condition in which memory or thinking difficulties are greater than expected for age but are not yet severe enough to prevent normal daily activities [3]. Importantly, MCI does not always progress to dementia; some individuals remain stable and some may even improve [3]

Among patients who do decline, the speed and pattern of decline can differ substantially. A longitudinal study of 936 individuals with MCI from the Alzheimer's Disease Neuroimaging Initiative (ADNI), for example, identified four distinct cognitive trajectories ranging from almost no decline to aggressive decline [4]. This means that two patients who appear similar during one clinical visit may follow very different paths in the following months or years.

This variability creates the central problem addressed by this project: a single clinical visit can describe a patient's current condition, but it cannot by itself show the direction or speed of that patient's cognitive change. Information relevant to Alzheimer's progression may be collected across several visits and may include cognitive and functional assessments, demographic characteristics, neuroimaging, genetic information, and biological markers. Looking at these measurements longitudinally allows changes and patterns to be captured rather than treating every examination as an isolated snapshot.

Evidence supports the value of this approach. In a study using ADNI data, Ding et al. compared a model based on several visits with a model based only on a single examination and found that incorporating longitudinal, multimodal information improved the prediction of progression from MCI to Alzheimer's disease [5]. ADNI is particularly suitable for this type of research because it is a longitudinal, multicenter study containing repeated clinical, cognitive, imaging, genetic, and biomarker measurements from participants at different stages of cognitive impairment [6].

Technological solutions that organize longitudinal patient information and provide individualized progression estimates can support more informed clinical monitoring and follow-up. This is relevant both globally and locally, including in Saudi Arabia, where healthcare professionals and families can benefit from accessible tools that support the monitoring of patients experiencing cognitive decline. In response to this need, REPLICA proposes a web-based platform that uses longitudinal patient information to predict future progression using the Clinical Dementia Rating – Sum of Boxes (CDR-SB). The platform provides clinicians with access to patient information, historical and predicted CDR-SB measurements, progression attention indicators, and follow-up information. It also supports communication between clinicians and caregivers through report sharing. The system is intended to support clinical interpretation and monitoring rather than replace professional medical judgment.

This document presents the REPLICA project, beginning with the problem being addressed, the proposed solution, product vision, roadmap, objectives, scope, and development resources. It then introduces the theoretical background and reviews related research and existing systems. Finally, it describes the system's intended users, requirements elicitation process, proposed architecture, use-case diagram, Product Backlog, and non-functional requirements.

1.1 The Problem

For patients with Mild Cognitive Impairment (MCI), one of the major challenges is that their future cognitive course cannot be understood from their current cognitive status alone. Patients who appear similar at one point in time may subsequently follow substantially different trajectories.

In a longitudinal study of 936 individuals with MCI from the Alzheimer's Disease Neuroimaging Initiative (ADNI), four distinct cognitive trajectories were identified over five years: approximately 27% showed almost no cognitive change, whereas other groups experienced mild, moderate, or aggressive decline [4]. This heterogeneity makes it difficult to determine from a single assessment how quickly an individual patient's cognitive and functional condition is changing or where that trajectory may lead.

Consider two patients with MCI who obtain the same Clinical Dementia Rating – Sum of Boxes (CDR-SB) score during their current assessment. Although their scores are identical at that visit, their previous measurements may reveal very different patterns. One patient's score may have remained relatively stable across previous visits, while the other patient's score may have progressively increased, indicating worsening cognitive and functional impairment. Treating their current scores as isolated observations would therefore hide an important difference between the two patients: the direction and rate of change over time.

The challenge is not simply the availability of patient information, but how to use measurements collected across time to estimate an individualized future progression trajectory. Relevant information may include repeated cognitive and functional assessments together with demographic, genetic, and imaging-derived measurements collected throughout follow-up. Evidence suggests that preserving this temporal information is valuable: Ding et al. found that a model incorporating longitudinal multimodal measurements from multiple visits achieved better prediction of MCI-to-Alzheimer's progression than a model based on a single examination [5].

Beyond estimating future progression, presenting patient information in a clear and accessible way is also important for clinical monitoring. Clinicians need to review changes across visits, recognize patients who may require greater attention, and maintain effective communication with caregivers. Without a centralized view of patient history, predicted progression, and follow-up information,

understanding an individual patient's condition over time and organizing continued monitoring becomes more challenging.

Therefore, the specific problem addressed by REPLICA is the limited ability of a single-timepoint assessment to represent and anticipate an individual MCI patient's cognitive and functional change over time. The project focuses on transforming longitudinal patient information into a personalized estimate of future CDR-SB progression, while supporting clinicians in interpreting patient-specific changes, identifying patients requiring closer attention, and managing follow-up and communication with caregivers.

1.2 The Solution

REPLICA attempts to solve this problem through a web-based platform that follows each patient's cognitive and functional condition across multiple visits. The platform combines the patient's previous CDR-SB scores with relevant personal and clinical information to estimate future CDR-SB progression and present the results in a clear and accessible format for clinical monitoring.

The system will use longitudinal data from ADNI. Selected variables may include the actual month of each visit, CDR-SB scores, age, sex, education, APOE4 status, and selected numerical brain measurements such as hippocampal volume [6]. The project will use measurements already extracted from MRI scans rather than processing raw MRI images.

A Linear Mixed-Effects Model (LMM) will form the main prediction model. LMM is a statistical model designed for repeated measurements, allowing it to capture both fixed effects, which represent general patterns shared across the population, and random effects, which represent individual differences between patients. In REPLICA, it will learn the general pattern of CDR-SB change across the study population while also estimating patient-specific progression based on the visits available for each patient. Because time is entered as an actual numerical value, the model can work with irregularly spaced visits [7].

Personalization will become stronger as additional visits become available. A Gradient Boosting model, such as XGBoost [8], will then be used to learn and correct remaining prediction errors using relevant patient characteristics and the prediction produced by the mixed-effects model. The final output will include an estimated future CDR-SB score and a personalized progression trajectory [9], allowing clinicians to compare the patient's historical, current, and predicted condition.

The REPLICA platform will provide a centralized clinical dashboard through which clinicians can access patient profiles and review individual patient information. The Patient Overview will present relevant demographic and clinical information, while the CDR Overview will display historical, current, and predicted CDR-SB measurements to help clinicians understand how the patient's condition has changed across visits and how it is expected to progress.

The system will also provide a Risk Indicator that presents a progression attention indication based on the patient's predicted progression and available clinical history. This information will support clinicians in identifying patients who may require closer monitoring. A Follow-Up List will allow clinicians to organize and review patients requiring continued attention.

To support communication between clinicians and caregivers, REPLICA will allow clinicians to prepare patient reports and share relevant information with caregivers. Caregivers will have access to clinician-shared reports, educational information, recommended cognitive activities, and supportive features intended for their role, without requiring access to detailed clinical monitoring functions.

Because the project uses retrospective rather than live hospital data, its prediction and monitoring functionalities will be demonstrated using historical patient records. The platform is intended to demonstrate how longitudinal patient information can be used to support clinical monitoring and follow-up without requiring integration with live hospital systems.

The main benefit of REPLICA is that it converts separate patient assessments into one understandable, patient-specific view of cognitive and functional progression. It can help clinicians recognize gradual changes, compare patients with similar current conditions but different histories, identify patients who may require closer follow-up, and communicate relevant information to caregivers through reports. By combining prediction, visualization, follow-up management, and caregiver communication in one platform, REPLICA aims to make longitudinal patient information more accessible and useful for clinical monitoring.

REPLICA is intended to support clinician interpretation and follow-up. It will not autonomously diagnose Alzheimer's disease, make treatment decisions, or replace professional clinical judgment.

1.3 Product

1.3.1 Product Vision

For clinicians/ neurologists, and caregivers monitoring / supporting patients with Mild Cognitive Impairment (MCI). Who need an early, personalized indication of how a patient's condition is likely to progress, rather than only a snapshot of where they stand today. REPLICA is a patient-specific cognitive progression platform. That uses longitudinal clinical history, CDR-SB assessments, and relevant patient characteristics to identify patient-specific patterns of decline and predict likely future trajectories, while also providing caregivers with clinician-shared reports, educational content, and supportive cognitive activities. Unlike approaches based on single-visit assessments, which may not capture changes observed across repeated visits [5]. Our product combines patient-specific prediction with an evolving longitudinal representation, adapting to each patient's own pattern of change through their visit history.

1.3.2 Product Roadmap

Sprint	Target date	Planned work
Sprint 0	27 September	Requirements analysis; ADNI dataset access request; literature review; initial system design
Sprint 1	22 October	Data cleaning and merging pipeline, feature engineering, baseline model development
Sprint 2	19 November	Personalized prediction model development. clinician dashboard
Release 1		
Sprint 3	TBA	Caregiver dashboard design and development
Sprint 4	TBA	Gradient boosting correction layer; clinician-caregiver data integration
Sprint 5	TBA	Full system integration; testing; final report
Release 2		

Table 1: Product Road Map

1.3.3 Objectives

The objectives of this project are defined across three levels (product, project, and learning) to capture not only what the system aims to achieve for its users, but also the development process required to build it and the technical growth gained by completing it.

1.3.3.1 Product Objectives (Customer Focus – Plan)

1. To estimate an individual MCI patient's future CDR-SB trajectory and expected rate of cognitive decline, addressing the limitation of conventional single-timepoint assessment, which primarily reflects a patient's current state rather than anticipating how it may change over time. Clinicians benefit from earlier, more individualized awareness of a patient's likely progression.
2. To display each patient's historical and predicted CDR-SB trajectory alongside their personal information and clinical history, giving clinicians a single view of a patient's full record and cognitive course over time.
3. To provide a risk indicator that flags patients who may require greater clinical attention based on their predicted rate of cognitive decline.
4. To present model outputs through a clear clinical dashboard, solving the problem that raw AI/ML outputs are not directly usable by clinicians without data-science expertise.
5. To allow clinicians to send a notification to a patient's linked caregiver when that patient is flagged as needing greater attention, enabling timely communication between the clinical and caregiving sides.
6. To provide a caregiver-facing dashboard where caregivers can view the patient's information, receive notifications sent by the clinician, and access recommended cognitive activities to support the patient day-to-day.
7. To allow clinicians to write a report or note to a patient's caregiver, giving caregivers clear, clinician-authored context and guidance.

1.3.3.2 Project Objectives (Solution Focus – Plan)

1. To acquire and preprocess the ADNI/ADNIMERGE2 dataset, linking diagnosis, cognitive, genetic, and MRI-derived tables.
2. To construct longitudinal per-patient sequences (CDR-SB, MMSE, visit timing) with a patient-level train/test split to avoid data leakage.
3. To apply a Linear Mixed Model (LMM) to model longitudinal changes in patients' CDR-SB scores while accounting for both population-level trends and individual patient differences.
4. To extract meaningful longitudinal features from the LMM, such as estimated rate of change and patient-specific trajectory, for use in a downstream prediction model.
5. To train and validate a Gradient Boosting model using baseline variables, LMM-derived features, and LMM predictions to correct residual error and improve future CDR-SB trajectory forecasting.
6. To derive a risk indicator from the predicted CDR-SB trajectory (e.g., categorizing predicted decline into risk levels) to support clinical prioritization.
7. To evaluate the models using regression metrics (MAE, RMSE, R^2) with patient-level and cross-validation, comparing the baseline against the proposed LMM + Gradient Boosting pipeline.
8. To design and implement a clinician dashboard displaying patient information, historical and predicted CDR-SB trajectories, and risk indicators.
9. To implement a notification mechanism allowing clinicians to notify a patient's linked caregiver account when a patient is flagged as high risk.
10. To design and implement a caregiver dashboard displaying patient information, clinician notification, and recommended cognitive activities.
11. To curate a set of cognitive activities recommendations suitable for MCI/early cognitive decline and integrate them into the caregiver dashboard.

1.3.3.3 Learning Objectives (Student Focus)

1. To gain hands-on experience applying Linear Mixed Models to biomedical longitudinal data and understanding how statistical models represent changes in patient measurements over time.
2. To learn how to work with large, multi-table clinical datasets, linking records by patient and visit identifiers while maintaining data consistency.
3. To learn multimodal feature integration techniques for combining clinical, cognitive, genetic, and imaging-derived features for Alzheimer's progression prediction.
4. To learn how to extract and interpret individual longitudinal trajectories from Linear Mixed Models, including population-level effects, patient-specific effects, and rates of change.
5. To gain practical experience handling missing data, and irregular patient follow-up in medical ML.
6. To gain practical experience training and tuning Gradient Boosting models for clinical prediction and understanding their advantages for nonlinear relationships in tabular medical data.
7. To learn rigorous model validation techniques, including patient-level splitting, cross-validation, prevention of data leakage, and comparison against baseline models.
8. To learn how to design a multi-role system with separate clinician- and caregiver-facing interfaces, each tailored to a different user's needs and technical background.
9. To learn how to design an interactive dashboard that presents AI/ML predictions, trajectories, and risk indicators to non-technical clinical and caregiving users.
10. To learn how to design a simple notification mechanism connecting two distinct user roles within the same system.

1.3.4 Scope

REPLICA is a web-based platform designed to support the longitudinal monitoring of patients with Mild Cognitive Impairment (MCI) by forecasting their future cognitive and functional progression. The project focuses on producing a personalized CDR-SB trajectory from longitudinal patient data and presenting the results through separate clinician and caregiver interfaces.

The scope of the project is defined as follows:

1. Platform:
   REPLICA will be developed as a web-based system that can be accessed through a standard web browser. A native mobile application for iOS or Android is not included in the current project scope.
2. Language:
   The system interface and its content will be provided in English only. Arabic localization and multilingual support are not included in the current version.
3. Target Users:
   The system will support two main user roles: clinicians and caregivers. Patients will not have a separate user account or patient-facing dashboard in the current version.
4. Data Source:
   Model development will use retrospective longitudinal data from the Alzheimer's Disease Neuroimaging Initiative (ADNI). The project will not collect new patient data directly from hospitals, clinics, or real patients for model training.
5. Clinical Data Input:
   Patient records used for prediction will be derived from the prepared ADNI dataset. Direct integration with Electronic Health Records (EHRs), hospital databases, or other live clinical systems is outside the current scope.
6. Prediction Target:
   The primary predictive outcome will be the Clinical Dementia Rating–Sum of Boxes (CDR-SB). REPLICA will estimate future CDR-SB values and present them as a personalized progression trajectory. The system will not independently diagnose Alzheimer's disease.

7. Patient Variables:

   The prediction models may use longitudinal CDR-SB measurements together with selected variables such as visit timing, age, sex, education, APOE4 status, and numerical brain measurements such as hippocampal volume, depending on their availability and suitability in the ADNI dataset.

8. Imaging:

   REPLICA may use numerical measurements already extracted from MRI data. Processing, segmenting, or interpreting raw MRI images is not included in the project.

9. Prediction Models:

   A Linear Mixed-Effects Model (LMM) will be used as the main longitudinal model. A Gradient Boosting model such as XGBoost will be evaluated as a correction layer to determine whether it improves prediction performance. Deep-learning models and large-scale GPU-based training are not part of the current scope.

10. Model Evaluation:

    Prediction performance will be evaluated using regression metrics such as MAE, RMSE, and R^2, with patient-level data splitting and cross-validation where appropriate. The proposed approach will also be compared with a simpler baseline model.

11. Prediction Uncertainty:

    The system will present prediction uncertainty where supported by the final modeling approach. Predictions will be presented as estimates rather than guaranteed future outcomes.

12. Clinician Interface:

    The clinician dashboard will allow clinicians to review patient information, historical CDR-SB measurements, predicted future CDR-SB trajectories, prediction uncertainty, and relevant follow-up information. Clinicians will also be able to prepare reports and send notifications for caregivers.

13. Caregiver Interface:

    The caregiver dashboard will provide clinician-shared reports, educational information about MCI and Alzheimer's disease, supportive cognitive activities, and caregiving resources. Caregivers will not have direct access to raw prediction results, detailed model outputs, or the clinician's full clinical dashboard.

14. Hardware:

    The project does not require wearable devices, medical sensors, IoT devices, or other dedicated hardware. Clinicians and caregivers will access the system using standard devices capable of running a web browser.

15. Real-Time Monitoring:

    Continuous or real-time patient monitoring is not included. The project uses longitudinal clinical observations rather than continuously streamed physiological data.

16. Medical Decisions:

    The system is intended to support clinical interpretation and follow-up. It will not autonomously diagnose patients, prescribe medication, recommend medical treatment, or replace professional clinical judgment.

17. Population Generalizability:

    The prediction models are developed using ADNI data and are not clinically validated on the Saudi population. Therefore, their generalizability to local populations cannot be assumed.

Overall, REPLICA is limited to demonstrating the feasibility of a web-based, English-language personalized cognitive progression system that uses longitudinal ADNI data to forecast future CDR-SB trajectories and present them through clinician- and caregiver-facing interfaces. Native mobile applications, Arabic localization, live hospital integration, wearable monitoring, raw medical-image processing, autonomous diagnosis, and real-world clinical deployment are outside the scope of the current project.

1.3.5 Hardware/Software Tools and Cost

Hardware Tools	
Personal Laptops (team members)	0 SAR (already owned)
Google Colab (cloud GPU)	0SAR
Google Drive (15 GB free)	0SAR
Software Tools	
Jupyter Notebook	0 SAR
Git + GitHub	0 SAR
Visual Studio Code	0 SAR
Jira	0 SAR
Python	0 SAR
Pandas and NumPy	0 SAR
Statsmodels	0 SAR
Scikit-learn, XGBoost	0 SAR
ADNI data access	0 SAR (requires approved research data-use application)

Table 2: Tools and Cost

1.4 Scrum Team

1.4.1 Skill Set Requirements

Technical Skill Required	Level & Learning Plan
Python programming	Intermediate
Data Preprocessing and Integration	Intermediate
Longitudinal Data Modeling	Beginner Learning plan: Online courses
Linear Mixed-Effects Models (LMM)	Beginner
Gradient Boosting / XGBoost	Beginner
Machine Learning Evaluation and Validation	Intermediate
Clinical and Multimodal Data Interpretation	Beginner
Web Development	Intermediate
Database Management	Intermediate
Git and GitHub Version Control	Intermediate
Agile / Scrum methodology	Intermediate
Alzheimer's clinical-data interpretation	Beginner Conduct a literature review and consult domain experts

Table 3: Skill Set Requirements

1.4.1.1 Learning

The learning required for REPLICA focuses on both the technical methods used in the system and the clinical context of Alzheimer's progression. The team has identified the main knowledge gaps and defined learning activities to address them throughout the project.

To strengthen the required technical knowledge, the team has begun reviewing official documentation, online courses, and tutorials for methods that are currently less familiar, particularly Linear Mixed-Effects Models (LMM), longitudinal data modeling, and Gradient Boosting/XGBoost. The project also provides practical learning through working with longitudinal clinical data, including preprocessing and integrating patient records across multiple visits, handling missing values and irregular follow-up times, and maintaining patient-level separation during model training and evaluation. These activities support the project's learning objectives related to longitudinal modeling, multimodal data integration, prediction, and model validation.

Learning is also required to understand the clinical meaning of the data used by the system. Since the team currently has limited experience with Alzheimer's clinical-data interpretation, the team has conducted a literature review and consulted domain experts, including clinicians and a caregiver, through requirements interviews to improve its understanding of Mild Cognitive Impairment (MCI), Clinical Dementia Rating – Sum of Boxes (CDR-SB), cognitive and functional decline, caregiver needs, and the clinical variables used in the ADNI dataset.

The team will also develop its understanding of how clinical information and predictive results should be presented through a healthcare-oriented interface. This includes designing clear visualizations for historical, current, and predicted CDR-SB information, presenting progression attention indicators in an understandable manner, supporting patient follow-up, and implementing appropriate clinician-caregiver report sharing and caregiver support features. The team will also strengthen its practical knowledge of Streamlit to develop the system's interactive interface.

At the current stage, the team has an intermediate level in Python programming, data preprocessing and integration, machine learning evaluation and validation, general web development, version control, and Agile/Scrum methodology. Database management is currently at a beginner-to-intermediate level. The team is still at a beginner level in longitudinal data modeling, Linear Mixed-Effects Models, Gradient Boosting/XGBoost, Streamlit, and clinical and multimodal Alzheimer's data

interpretation. These areas will continue to be developed progressively as the project moves from data preparation and baseline modeling toward CDR-SB progression prediction, progression attention indication, clinician-caregiver report sharing and caregiver support, and full dashboard implementation.

1.4.2 Roles and Responsibilities

Scrum Team	
Product Owner:	Dr. Sharefah Alghamdi
Developers:	Reem Almaskati Sadeem Al Sultan Sondos Aldughaim Danah AlDhowaihy
Scrum Master (SM):	Sondos Aldughaim
Stakeholders:	Project Supervisor- Physicians/clinicians - Patients with MCI- Saudi Ministry of Health- Caregivers - King Saud Univesity- Team members

Table 4: Roles and Responsibilities

2 Background

REPLICA is built around the idea of following how a patient's cognitive and functional condition changes over time. To represent this change in a measurable way, the project uses the Clinical Dementia Rating–Sum of Boxes (CDR-SB) as its primary progression measure. The Clinical Dementia Rating (CDR) evaluates six areas that may be affected by cognitive decline: memory, orientation, judgment and problem solving, community affairs, home and hobbies, and personal care [10]. Each area is scored separately, and the scores can be added together to produce the CDR-SB, which ranges from 0 to 18. Higher scores indicate greater impairment [10], [11]. Unlike an assessment that focuses only on cognitive test performance, CDR-SB reflects both cognitive ability and its effect on everyday functioning, which has made it a commonly used outcome for measuring changes in Alzheimer's disease and related cognitive impairment over time [11], [12].

A single CDR-SB value, however, describes only one point in a patient's history. A score of 3, for example, does not by itself reveal whether the patient has remained close to that level for a long period or has recently changed from a much lower score. This becomes clearer when several observations from the same patient are placed in chronological order. A patient might have a CDR-SB of 1.0 at the first visit, 1.5 several months later, 2.5 at another visit, and 4.0 at a later stage. Together, these measurements form a trajectory, showing the direction and rate of change rather than only the patient's latest condition. Previous longitudinal research has used CDR-SB specifically to study rates of Alzheimer's disease progression, supporting its use as a measure that can be followed across time [12].

Data collected from the same person repeatedly over time is known as longitudinal data. This differs from cross-sectional data, where an individual is observed at only one point in time. Longitudinal data is especially useful when the objective is to understand change because it connects each new observation to what was previously known about the same individual. At the same time, it introduces an important statistical consideration: repeated measurements from one patient are related to each other and should not be treated as completely independent observations. Methods developed for longitudinal analysis therefore need to account for the relationship between measurements belonging to the same individual [7].

Another characteristic of clinical follow-up is that visits do not always occur at equal intervals. A patient may be assessed at baseline, return six months later, and then attend the next visit eight or

twelve months after that. For this reason, simply labeling the records as Visit 1, Visit 2, and Visit 3 can hide useful information about how much time actually passed between measurements. REPLICA therefore uses the actual visit timing when constructing each patient's longitudinal history. This allows a change that occurred over six months to be distinguished from the same numerical change occurring over a much longer period.

To study this type of progression, the project requires a dataset containing repeated observations from the same individuals. REPLICA uses data from the Alzheimer's Disease Neuroimaging Initiative (ADNI), a longitudinal, multicenter research study containing clinical, cognitive, demographic, genetic, imaging, and biomarker information collected from participants across different stages of cognitive impairment [6]. Because patients may have several visits and several types of measurements, ADNI provides the type of longitudinal and multimodal information needed to study patient-specific change.

CDR-SB is the main outcome that REPLICA aims to forecast, but the model can use additional patient information to provide context for that trajectory. These variables may include age, sex, education, genetic information, and structural brain measurements available from ADNI. Other cognitive assessments such as MMSE (Mini-Mental State Examination) are also available in the dataset and may provide supporting information, but they are not the primary predicted outcome of the current system.

One of the genetic variables considered in the project is APOE4. The APOE gene exists in several forms, and the APOE ε4 allele is associated with an increased risk of Alzheimer's disease and, in some populations, an earlier age of onset [13]. However, APOE4 is a risk factor rather than a diagnosis. Carrying the APOE ε4 allele does not mean that a person will necessarily develop Alzheimer's disease, and the strength of its association with Alzheimer's risk can differ across populations [13]. Within REPLICA, APOE4 is therefore treated as one piece of patient information that may contribute to prediction rather than as a direct indicator of the patient's future condition.

Structural information from brain imaging can provide another source of context. One measurement considered in REPLICA is hippocampal volume. The hippocampus plays an important role in memory, and reductions in hippocampal volume have been studied extensively in relation to Alzheimer's disease and cognitive decline. MRI research has shown that hippocampal atrophy can be associated with disease progression, although hippocampal volume alone is not specific enough to diagnose

Alzheimer's disease [14]. REPLICA does not process raw MRI scans or attempt to identify abnormalities directly from medical images. Instead, it uses numerical structural measurements that have already been extracted from MRI data, such as hippocampal volume. This keeps the project focused on progression modeling rather than medical image processing.

Once these observations are organized across time, the next challenge is to model them in a way that recognizes both similarities and differences between patients. If one average progression line were applied to every patient, important individual variation could be lost. At the same time, building a completely separate model for every patient would be difficult, particularly when only a small number of visits are available. REPLICA addresses this balance using a Linear Mixed-Effects Model (LMM).

Linear Mixed-Effects Models are designed for data containing repeated observations from the same individuals. In simple terms, the model first learns the general pattern of change across all patients and then adjusts that pattern to better represent each individual patient [7].

The general pattern shared across the study population is represented by fixed effects. For example, the model may learn how CDR-SB typically changes over time and how selected patient characteristics are related to that change.

However, patients do not all begin at the same CDR-SB level or progress at the same rate. The model therefore uses random effects to represent these individual differences. A random intercept allows each patient to have a different starting level, while a random slope allows each patient to have a different rate of change over time. In this way, REPLICA can begin with what is learned from the wider population and then adapt the predicted trajectory according to the individual patient's available history.

This can be summarized as:

Population pattern + Patient-specific information → Personalized progression trajectory

The amount of patient-specific information available naturally depends on the patient's history. When only limited observations are available, there is less direct evidence about that individual's rate of change, so the estimate depends more heavily on the patterns learned from the wider population and the patient characteristics available at that stage. As additional visits are recorded, more information about the patient's own longitudinal pattern becomes available and can be incorporated into later

predictions. This is why repeated follow-up is important to the personalization approach used in REPLICA.

Although an LMM provides a useful structure for repeated patient measurements, it represents relationships according to a specified statistical form and may not capture every complex relationship among the available variables. Factors such as age, education, genetic information, imaging-derived measurements, and previous clinical scores may interact in ways that are not fully represented by the longitudinal model alone. For this reason, REPLICA also investigates Gradient Boosting as an additional correction layer.

Gradient Boosting is a machine-learning approach that combines multiple decision trees sequentially. Instead of building all trees independently, each new tree focuses on improving the errors remaining from the previous predictions. XGBoost is a widely used implementation of tree boosting designed for efficient and scalable predictive modeling [8]. In REPLICA, Gradient Boosting is not intended to replace the longitudinal model. Instead, the proposed approach allows the LMM to provide the longitudinal foundation and patient-specific progression information, while the Gradient Boosting layer learns additional patterns from variables such as patient characteristics, imaging-derived measurements, and the output of the LMM.

Conceptually, the process can therefore be viewed as two connected stages. The LMM first models the structure of the patient's repeated observations and estimates a personalized progression pattern. Gradient Boosting can then examine the remaining prediction error and determine whether additional nonlinear relationships can improve the final estimate. The performance of this combined approach must be evaluated against simpler baseline models to determine whether the additional complexity improves forecasting rather than assuming that it will.

The final predictive result is a future CDR-SB trajectory. Rather than asking only what category a patient belongs to, the system estimates how the patient’s CDR-SB may change at selected future time points. Historical measurements can therefore be displayed alongside predicted values, allowing the clinician to see where the patient has been, where the patient is currently, and the direction suggested by the model for future follow-up.

A prediction, however, should not be interpreted as a certain future outcome. Medical data contains variability; patients do not always follow the patterns observed in the training population, and every

statistical prediction contains uncertainty. A predicted CDR-SB value should therefore be understood as an estimate based on the information currently available. Where possible, the interface can communicate this uncertainty rather than presenting the prediction as an exact value that is guaranteed to occur. This is particularly important in REPLICA because the purpose of the system is to support clinical understanding, not to make autonomous decisions.

The system may also summarize predicted change using a simple progression attention indicator to make the information easier to review. Such an indicator must be distinguished from a diagnosis or from a calculated probability that the patient will develop Alzheimer's disease. Unless a separate validated classification model is developed for that purpose, the indicator in REPLICA represents the level of attention suggested by the predicted CDR-SB trajectory, for example according to the expected direction or rate of change. The detailed trajectory remains available so that the clinician can see the information behind the simplified indicator.

Presenting the prediction clearly is as important as producing it. Clinical Decision Support systems are intended to provide clinicians with relevant patient-specific information that can help inform care decisions rather than replace professional judgment [15]. REPLICA follows this principle by presenting the patient's historical measurements, predicted progression, and related information through a web-based clinician interface. The clinician remains responsible for interpreting the prediction together with other clinical information and deciding whether any follow-up action is appropriate.

The caregiver-facing side of REPLICA has a different purpose from the clinician-facing component. Detailed clinical information, predicted trajectories, and model outputs remain available to healthcare professionals, while the caregiver interface focuses on education, day-to-day support, and clinician communication. Caregivers can receive notifications and reports prepared and shared by the clinician, allowing important patient information to be communicated in a controlled and understandable form without exposing raw prediction results. The caregiver interface can also provide clear educational information about Mild Cognitive Impairment and Alzheimer's disease, recommended cognitive activities that can be performed with the patient, and reminders to help organize caregiving tasks. The information is presented in a simple and accessible form because caregivers are not expected to have a medical or technical background. This keeps clinical interpretation with the healthcare professional while still giving caregivers useful information and supportive resources.

As a result, REPLICA can be viewed as a connected longitudinal workflow rather than only a prediction algorithm. A patient's available clinical visits contribute CDR-SB measurements and other relevant information to their longitudinal history. The longitudinal model uses this record to estimate the patient-specific pattern, the prediction pipeline produces a future trajectory, and the web application presents the result alongside the patient's previous observations. As additional eligible visits become available, the history can be extended and the trajectory recalculated using the new information. Since the current project uses retrospective ADNI records, this process will be demonstrated using historical patient data rather than live clinical monitoring.

The software required to support this process is primarily Python-based. Statistical modeling can be performed using libraries that support mixed-effects models, while XGBoost and Scikit-learn can support the machine-learning and evaluation components. The web interface can then connect the model outputs with the clinician- and caregiver-facing functions of the system. No dedicated medical sensors or wearable hardware are required for the current project because the system works with longitudinal clinical data rather than continuous real-time physiological streams.

Together, these concepts provide the technical foundation for REPLICA. CDR-SB supplies a measurable representation of cognitive and functional progression, longitudinal data shows how that measurement changes across visits, ADNI provides the patient histories and related variables required for model development, LMM allows population patterns and patient-specific differences to be represented together, and Gradient Boosting provides an additional method for capturing relationships that may remain after longitudinal modeling. The resulting predictions are then presented as clinical decision-support information through a system designed to help clinicians follow patient progression over time while maintaining a clear distinction between computational prediction and professional clinical judgment.

3 Literature Review

This section reviews previous research that provides the scientific foundation for REPLICA. The selected studies address several related aspects of cognitive progression in Mild Cognitive Impairment (MCI), including differences in progression between patients, the value of longitudinal patient history, the prediction of future cognitive change, patient-specific trajectory modeling, and the use of digital technologies to support patients and caregivers. Together, these studies help explain why REPLICA focuses on longitudinal patient information, personalized cognitive progression modeling, forecasting future CDR-SB trajectories, and supporting both clinical monitoring and caregiver involvement.

Research on Alzheimer's disease increasingly considers not only a patient's current cognitive state, but also how that state may change over time. This is particularly important for individuals with MCI because patients who appear similar at one point in time may subsequently follow different cognitive trajectories.

Wang et al. [4] demonstrated this heterogeneity using longitudinal ADNI data from 936 individuals with MCI. Rather than identifying a single common pattern of cognitive decline, the researchers identified four distinct cognitive trajectories, ranging from minimal decline to more aggressive deterioration. These findings indicate that cognitive progression among individuals with MCI is heterogeneous and that patients may experience substantially different patterns of change over time.

This variability creates an important challenge for prediction. If patients with similar cognitive states at one visit can subsequently follow different trajectories, an assessment of the current state alone may not fully represent how their conditions are developing. Longitudinal information can provide additional context by showing how a patient has changed across previous visits.

Zhang and Shen [16] investigated future clinical prediction using longitudinal and multimodal information from ADNI. Their work used measurements from previous visits to predict future cognitive outcomes, including MMSE. Although REPLICA uses CDR-SB rather than MMSE as its primary progression measure, the study remains relevant because it demonstrates the broader value of using previous patient observations to predict future cognitive change rather than focusing exclusively on the patient's current state.

This distinction is important for REPLICA. The objective is not simply to determine whether a patient may eventually progress to Alzheimer's disease, but to estimate how the patient's cognitive and functional condition may evolve over time. Repeated observations therefore become important because they can reveal whether a patient has remained relatively stable or has experienced a pattern of progressive change.

Ding et al. [5] further investigated the value of longitudinal information using multimodal ADNI data to predict progression from MCI to Alzheimer's disease. Their approach incorporated measurements collected across multiple visits and achieved stronger predictive performance than an approach based on a single examination. Beyond the reported performance difference, this study demonstrates that longitudinal observations contain temporal information that can be lost when a patient is represented using only a single visit.

However, incorporating multiple visits introduces another challenge. Patients may differ in their initial cognitive status, their rates of change, and the timing of their follow-up visits. A longitudinal prediction approach should therefore account not only for overall patterns within the population but also for differences between individual patients.

Platero and Tobar [17] addressed longitudinal patient variation using Linear Mixed-Effects Models with repeated clinical and MRI-derived measurements from ADNI. Their work illustrates how longitudinal information can be modeled while accounting for individual differences and reports improved prediction as additional longitudinal information becomes available. This approach is particularly relevant to REPLICA because mixed-effects modeling can represent population-level patterns while also allowing patient-specific differences in longitudinal progression.

For REPLICA, this means that a patient's previous visits are not treated merely as historical records. Instead, the longitudinal history contributes to the estimation of that patient's progression. As additional observations become available, the amount of patient-specific information available to the model increases, supporting a more individualized representation of cognitive change.

Longitudinal prediction can also extend beyond predicting a single future outcome. Cognitive progression occurs across time, making it useful to consider how clinical measurements may evolve across several future time points rather than estimating only one endpoint.

The TADPOLE Challenge [18] reflects this forecasting perspective. Researchers used historical ADNI information to forecast future Alzheimer's-related outcomes, including cognitive measures and brain measurements. The challenge demonstrated that forecasting future clinical evolution is an important prediction problem and that different types of outcomes may benefit from different modeling approaches. This is relevant to REPLICA because the system is intended to represent expected cognitive progression across time rather than provide only an isolated future classification.

More directly related to REPLICA's current prediction target, Jamalian et al. [9] investigated the longitudinal trajectory of the Clinical Dementia Rating–Sum of Boxes (CDR-SB). Their work modeled disease progression over time while accounting for differences in the rates at which individual patients changed. The use of CDR-SB as a longitudinal outcome is particularly relevant to REPLICA because CDR-SB is also used as the primary measure for representing changes in cognitive and functional progression over time.

The concept of a trajectory is important because a patient's future condition cannot always be adequately represented by a single predicted value. A trajectory provides a broader representation of expected progression by describing how CDR-SB may change across future time points. This allows the system to represent whether the predicted course remains relatively stable or shows a pattern of increasing impairment over time.

Another important characteristic of longitudinal monitoring is that the available information about a patient changes over time. A prediction produced from early visits does not need to remain fixed after additional observations become available. A newly observed CDR-SB measurement provides additional information about the patient's actual progression and can therefore be incorporated into subsequent predictions. In REPLICA, patient visits are processed chronologically so that the patient-specific prediction can be recalculated as additional longitudinal information becomes available. This allows the predicted trajectory to reflect the patient information available at each stage of follow-up.

While mixed-effects models provide a structured way to represent longitudinal and patient-specific variation, clinical progression may also involve nonlinear relationships that are not fully represented by a linear model. Olaniran et al. [19] investigated this broader methodological challenge by combining mixed-effects modeling with Gradient Boosting for longitudinal data. Their approach uses mixed-effects components to account for subject-specific variability and within-subject correlations,

while Gradient Boosting provides flexibility for modeling nonlinear relationships and interactions. Although their proposed framework differs from the architecture used in REPLICA, the study demonstrates the methodological value of combining longitudinal mixed-effects modeling with Gradient Boosting when both subject-specific variation and nonlinear patterns need to be considered.

This provides support for the complementary modeling strategy proposed for REPLICA. The Linear Mixed-Effects Model serves as the primary longitudinal model for representing population-level progression and patient-specific differences. A Gradient Boosting correction layer is then investigated to learn residual patterns that may not be fully captured by the linear longitudinal structure. Rather than replacing the longitudinal model, the correction layer complements it by providing additional flexibility for relationships that may be nonlinear.

The clinical prediction component represents one side of the system, but cognitive decline also extends beyond clinical visits. People living with MCI or dementia often receive support outside clinical settings, making digital technologies potentially relevant not only for clinicians but also for patients and those involved in their care.

Di Lorito et al. [20] conducted a systematic review and meta-analysis of digital health interventions for people living with dementia and MCI. The review included 20 studies covering different types of digital interventions and reported outcomes related to cognitive, physical, behavioral, psychological, and daily functioning. The authors also emphasized that accessibility, acceptability, and sustainability for end users are important considerations when developing digital health services.

These findings are relevant to the caregiver-facing component of REPLICA. While the clinician-facing component focuses on reviewing longitudinal patient information and predicted cognitive progression, the caregiver-facing component extends the platform beyond the clinical setting by providing digital support for individuals involved in the patient's day-to-day care. This component is not intended to replace professional clinical assessment. Instead, it provides a complementary interface through which caregivers can access clinician-shared reports, educational information, recommended cognitive activities, and supportive resources.

When these studies are considered together, they provide a connected foundation for the main design decisions of REPLICA. Patients with MCI can follow heterogeneous cognitive trajectories [4], and previous observations can be used to predict future cognitive outcomes [16] . Longitudinal patient

history can provide predictive information beyond that available from a single examination [5], while mixed-effects modeling provides an approach for representing both population-level patterns and differences between individual patients [17]. Forecasting research demonstrates the value of representing future clinical change across time [18], and longitudinal CDR-SB modeling provides direct support for using CDR-SB trajectories to characterize progression [9]. Research combining mixed-effects modeling with Gradient Boosting further demonstrates how structured longitudinal modeling can be complemented by flexible nonlinear modeling [19], while research on digital health interventions demonstrates the relevance of digital support for people with cognitive impairment and those involved in their care [20].

Together, these findings shape the requirements of REPLICA. The system should preserve and use information from previous patient visits rather than relying only on the most recent examination. It should account for the timing of repeated observations and differences between individual patients. It should combine longitudinal cognitive information with relevant patient characteristics and represent expected cognitive progression as a future CDR-SB trajectory rather than only as a single disease classification. The prediction should also be capable of being recalculated as additional visits become available.

These requirements support the proposed modeling approach. A Linear Mixed-Effects Model provides the longitudinal foundation by representing population-level progression while accounting for patient-specific differences. A Gradient Boosting model is evaluated as a correction layer for residual patterns that may not be fully represented by the linear longitudinal model. The resulting historical and predicted progression is presented through a clinician-facing interface, allowing clinicians to review patient information, predicted CDR-SB trajectories, progression attention indicators, and follow-up requirements.

A separate caregiver-facing component extends the platform to support the patient's care outside the clinical setting. Through this interface, caregivers can receive reports and notifications shared by clinicians and access educational information and supportive cognitive activities. This distinction allows detailed clinical interpretation and progression monitoring to remain with healthcare professionals while providing caregivers with understandable information relevant to their responsibilities.

REPLICA therefore brings together several related areas of previous research: heterogeneous progression among patients with MCI, the value of longitudinal history, patient-specific progression modeling, CDR-SB trajectory forecasting, complementary nonlinear modeling, and digital support for cognitive impairment. Rather than relying only on a snapshot of the patient's current condition, the system uses the patient's longitudinal history to estimate how their cognitive and functional condition may change over time and allows that estimate to be revised as additional information becomes available.

3.1 Competitive Product Analysis

Existing research models and clinical decision-support systems address different aspects of cognitive prognosis, longitudinal disease progression, and patient-specific prediction in Alzheimer's disease. Three relevant solutions were selected for comparison with REPLICA: CognitiveTwin, the Florey Fusion Model (FFM), and PredictND. Each solution represents a different approach to cognitive progression, including patient-specific longitudinal modeling, future cognitive forecasting, and clinical decision support. Together, these solutions provide relevant points of comparison across the technical, predictive, and clinical aspects addressed by REPLICA.

CognitiveTwin

CognitiveTwin is a research Digital Twin framework developed for patient-specific prediction of cognitive decline in Alzheimer's disease [21]. The framework is designed to model individual cognitive trajectories using longitudinal patient information and account for differences in disease progression between patients.

CognitiveTwin integrates longitudinal multimodal data from several sources, including cognitive assessments, magnetic resonance imaging (MRI), positron emission tomography (PET), cerebrospinal fluid biomarkers, and genetic information. It uses a Transformer-based architecture to integrate these modalities and a Deep Markov Model to represent temporal disease dynamics and generate patient-specific cognitive trajectories.

The framework was trained and evaluated using longitudinal data from 1,666 patients in the TADPOLE dataset derived from the Alzheimer's Disease Neuroimaging Initiative (ADNI). Its evaluation also examined demographic fairness and robustness to missing-not-at-random clinical data.

CognitiveTwin provides a relevant comparison for REPLICA because both approaches use longitudinal patient information to account for individual differences in cognitive progression. However, CognitiveTwin focuses on a research-oriented Digital Twin framework, while REPLICA proposes a web-based monitoring platform that combines CDR-SB trajectory prediction with clinician follow-up and caregiver support.

Florey Fusion Model (FFM)

The Florey Fusion Model is a research model developed by researchers at the Florey Institute of Neuroscience and Mental Health for predicting future cognitive outcomes in individuals with MCI or Alzheimer's disease [22]. The model uses patient information available at assessment to estimate future cognitive progression over a period of three years.

FFM uses a fusion framework that combines classification and regression approaches, with several machine-learning algorithms evaluated during its development. It generates predictions for multiple cognitive outcomes, including MMSE and CDR-SB scores.

In its development and validation study involving 238 participants, the model achieved an AUC-ROC of 0.91 for predicting conversion from MCI to Alzheimer's disease, with mean absolute errors of 1.51 for MMSE and 1.32 for CDR-SB. The model was also pilot-tested on an additional 30 participants through a web-based research tool to examine its performance when parts of the available patient information were missing.

FFM is particularly relevant to REPLICA because it demonstrates the use of machine learning to forecast future CDR-SB measurements. However, REPLICA places additional emphasis on using repeated patient observations to estimate individualized longitudinal trajectories and presenting these predictions alongside patient history, progression attention indicators, and follow-up information.

PredictND

PredictND is a clinical decision-support tool developed for use in memory clinics [23]. It is designed to assist clinicians in assessing patients with subjective cognitive decline (SCD) or Mild Cognitive Impairment (MCI) and estimating the likelihood of future disease progression.

PredictND integrates multiple types of patient information collected during the diagnostic process, including demographic characteristics, cognitive test results, cerebrospinal fluid biomarkers, and MRI findings. The system combines this information into a clinical overview and provides a classification describing the likelihood of disease progression.

PredictND was prospectively evaluated in a multicenter study involving 429 patients with SCD or MCI who were followed for at least one year. The evaluation found that using PredictND increased clinicians' confidence in their prognostic assessments, although it did not significantly improve overall prognostic accuracy across the full study population.

PredictND provides a relevant clinical comparison for REPLICA because both systems aim to present predictive information in a format that supports healthcare professionals. However, while PredictND primarily focuses on supporting clinical prognosis using baseline patient information, REPLICA focuses on forecasting patient-specific CDR-SB trajectories from longitudinal records and integrating these results with follow-up management and clinician-caregiver communication.

The table below provides a brief comparison of the selected competitors and REPLICA across key features and capabilities.

	CognitiveTwin	**FFM**	**PredictND**	Replica
Primary context	Cognitive progression research	Cognitive prognosis research	Memory-clinic decision support	Longitudinal clinical monitoring
Primary users	Researchers	Researchers	Clinicians	Clinicians and caregivers
Main predictive focus	Patient-specific cognitive trajectories	Future cognitive scores and MCI-to-AD progression	Likelihood of disease progression	Future patient-specific CDR-SB trajectory
Patient-specific prediction	Yes	Yes	Yes	Yes
Longitudinal info used for trajectory modeling	Yes	Not the primary longitudinal updating approach described	Progression evaluation based primarily on baseline information	Yes
Repeated-visit update workflow	clinical updating is not its primary presented workflow	Not described as a continuous updating workflow	Not described as a continuous updating workflow	Core proposed workflow
Clinician-facing monitoring interface	Not the primary focus of the research framework	Research web-based tool	Yes	Yes
Caregiver-facing functionality	Not described	Not described	Not described	Yes

Table 5: Competitive Product Analysis

Based on the competitive analysis, CognitiveTwin is the closest approach to REPLICA conceptually, particularly in its use of longitudinal patient data to model individual cognitive progression. FFM is more closely aligned with REPLICA's forecasting objective, particularly the prediction of future CDR-SB outcomes, while PredictND provides the closest comparison from a clinical perspective through its use of predictive information to support clinicians in memory-care settings. This shows that REPLICA overlaps with existing approaches in several important areas, although each competitor emphasizes a different aspect of the problem.

More broadly, the analysis indicates that the field is moving beyond simply assessing a patient's current cognitive condition toward using patient data to anticipate how that condition may change over time. However, predictive modeling, longitudinal patient analysis, and clinical support are often emphasized separately rather than as parts of a continuous follow-up process.

This suggests an opportunity for systems that connect prediction with ongoing patient monitoring and make updated information usable throughout follow-up.

Within this landscape, REPLICA is positioned around longitudinal patient-specific follow-up. It combines longitudinal patient information with future CDR-SB forecasting in a workflow where predictions can be updated as new visits become available, while supporting neurologist monitoring and selected caregiver interaction. Therefore, REPLICA's proposed contribution lies not in introducing each capability independently, but in integrating longitudinal prediction, and clinical follow-up within a single system.

4 System Description

This section identifies the system's primary users; clinicians and caregivers of MCI patients, describes the methods used to elicit requirements for REPLICA, outlines the rationale for selecting interviews over other methods such as questionnaires or focus groups, and summarizes the key findings that shaped the project's design decisions.

4.1 Users

This subsection describes the general characteristics of the users of the system, including their educational background, professional experience, and technical expertise.

The system is primarily designed for clinicians and neurologists involved in monitoring patients with Mild Cognitive Impairment (MCI). It also supports caregivers who receive relevant information and communication from healthcare professionals regarding the patients they care for. The system allows clinicians to review patient information, historical and current CDR-SB scores, predicted CDR-SB progression, risk indicators, and patients requiring follow-up. Clinicians can also prepare reports and share relevant reports to caregivers when needed.

The general characteristics of the users can be summarized as follows:

- The primary users are clinicians, neurologists professionals.
- Healthcare professionals are expected to have an educational or professional background in healthcare, particularly in neurology, geriatrics, cognitive assessment, or related clinical fields.
- Clinicians and neurologists should have knowledge of cognitive impairment and clinical assessment measures such as the Clinical Dementia Rating – Sum of Boxes (CDR-SB), allowing them to interpret a patient’s current condition and changes across previous visits.
- Clinicians and neurologists should be able to understand changes in a patient’s cognitive and functional measurements across multiple clinical visits and use this information when reviewing the patient’s progression.

- Clinicians and neurologists are expected to have basic technical skills, such as navigating a web-based system, selecting patient profiles, viewing dashboards, interpreting graphs, reviewing risk indicators, accessing follow-up information, and preparing reports.
- Clinicians and neurologists do not require expertise in artificial intelligence, machine learning, or statistical modeling, as REPLICA presents its predictions and patient-specific progression through understandable visualizations and clinical information.
- Caregivers are secondary users of the system.
- They may receive clinician-shared reports and access educational content, supportive cognitive activities, and other relevant resources.
- Caregivers are not required to have a medical or healthcare background, as the information available to them is intended to be presented in a clear and understandable form.
- Caregivers are expected to have basic technical skills, such as logging into a web-based system, accessing shared information, and viewing reports or notifications provided by the healthcare professional.
- Caregiver access is limited to the information and functions intended for their role. Clinical monitoring functions, such as reviewing detailed patient progression, predicted CDR-SB trajectories, risk indicators, and managing the follow-up list, remain the responsibility of authorized healthcare professionals.
- Both user groups should have general familiarity with web-based applications, while the interface should remain clear and user-friendly so that users can perform their required tasks without needing to interpret raw model outputs.

4.2 Requirements Elicitation

Requirements for REPLICA were gathered using two complementary methods: a review of published literature and interviews with representatives of the system's two intended user groups. The literature review established the clinical and technical foundation for the project by drawing on peer-reviewed studies of MCI-to-AD progression modeling and longitudinal, multimodal prediction approaches, as well as ADNI's own data documentation to understand what patient information is realistically available for modeling. To complement this theoretical foundation with practical, first-hand insight, interviews were then conducted with two physicians who regularly see MCI patients in clinical practice and one caregiver of a person experiencing mild memory issues. Each interview followed a semi-structured set of open-ended questions covering current workflow, use of historical patient data, and reactions to specific features under consideration for the dashboard, with room for follow-up where relevant. No questionnaires, focus groups, or workshops were used; the small number of distinct user roles and the exploratory nature of the questions made individual interviews more suitable for surfacing detailed, example-driven feedback than a group-based or survey-based method would have provided at this stage. Full interview transcripts are provided in the Appendix.

The clinicians interviews revealed a consistent picture of time-constrained clinical practice, with one clinician reporting roughly ten minutes per patient in a government clinic setting and the other reporting fifteen to twenty minutes in a comparatively less constrained setting; both nonetheless emphasized that little time exists for reviewing supplementary tools during the consultation itself, and both independently stated that any feature requiring manual data entry mid-visit, or requiring them to leave their existing system to check a separate platform, would not be used regardless of its predictive quality. Both physicians also converged, without prompting from one another, on the idea that the trend or trajectory of a patient's cognitive scores matters far more than any single measurement, that MRI is primarily useful at baseline to rule out non-Alzheimer's causes, and that genetic testing for APOE4 is rarely available to them in practice despite its relevance in the literature. These points directly support the project's decision to center the clinician dashboard on a CDR-SB trajectory rather than a single score, and indicate that MRI- and genotype-derived features should be treated as available-if-present baseline inputs rather than data expected at every visit.

Both clinicians also described the same underlying difficulty that motivates this project: patients with MCI can appear clinically identical at a first visit yet follow entirely different courses, some remaining

stable for years and others converting to Alzheimer's disease within about a year. They attributed part of this difficulty to real-world data inconsistencies rather than to the disease alone, citing irregular follow-up, missed and delayed appointments, and testing conditions that vary from visit to visit, and both noted that confounding factors such as education level and comorbidities such as diabetes, depression, or poor sleep can distort raw cognitive scores in ways that a simple threshold-based interpretation would miss. Both physicians indicated that multiple visits are required, generally around two to four visits, spanning roughly a year or more, before feeling confident forming a judgment about a patient's likely trajectory, and both stated firmly that they would not act on a prediction presented as a single, unexplained number. They specifically asked to see the margin of error or confidence behind a predicted value rather than a "clean number," citing concern that an unexplained figure creates false confidence, and both raised the population on which any model is trained as a relevant concern, noting that ADNI's predominantly Western cohort may not fully reflect the education levels and risk-factor profiles of their own patients. Related to this, both clinicians independently preferred a categorical progression attention indicator, such as Low, Moderate, or High attention over a specific predicted number, explaining that a precise figure invites disagreement from families while a category is faster to communicate and act on during a busy clinic day.

The caregiver interview surfaced with a different but complementary set of needs. The caregiver reported no structured method of tracking the patient's condition, relying instead on informal notes taken inconsistently on a phone, and described frequently forgetting to mention important changes by the time of the doctor's appointment, which occurs only once every couple of months. This finding supports the value of the clinician-shared reports and supportive resources as addressing a genuine, currently unmet need. The caregiver also reported having tried a generic brain-game application previously, which the patient found frustrating and abandoned, in contrast to simple, low-setup activities such as reviewing old photographs or describing familiar recipes, which were better received; when asked what would make a recommended activity trustworthy, the caregiver stated that a recommendation linked to what the doctor had already advised would be trusted far more than one generated independently by the application. This insight supports tying the caregiver dashboard's activity recommendations to the clinician guidance or clinician-shared report or notification rather than presenting them as a fully independent feature, and suggests that recommended activities should lean toward simple, low-effort formats rather than assuming a gamified or app-based approach will suit every patient. Finally, the caregiver independently asked for a simple visual indication of whether the

patient's condition was worsening or remaining stable over time, and for something that could be shown directly to the doctor rather than relying on memory during the appointment, reinforcing the value of providing caregivers with clear clinician-shared summaries that can support communication during follow-upand the value of the notification mechanism connecting the clinician and caregiver dashboards, this time from the caregiver's side of that relationship.

Taken together, the interview findings largely validated the design decisions already reflected in the project's objectives and scope, while also surfacing considerations not explicitly addressed beforehand: the possibility that ADNI's demographic composition may not generalize perfectly to a local patient population. This point is documented here as findings drawn directly from intended end users and are carried forward as acknowledged limitations in the Scope section of this proposal, rather than treated as fully resolved by the current project.

4.3 Architecture

REPLICA follows a three-tier layered web architecture consisting of the presentation, application, and data layers, supported by a separate prediction subsystem and an offline model-development pipeline sourced from ADNI. Figure 1 shows the high-level architecture. The layered architecture separates presentation, application logic, and data management into distinct responsibilities [24].

We chose a layered architecture because REPLICA's responsibilities separate naturally into presentation, application logic, and data management, supporting separation of concerns and making the system easier to maintain and extend.

System users: Clinicians and caregivers access REPLICA through a standard web browser. No dedicated external medical device or sensor is required.

Major components: the Presentation Layer, which contains the clinician and caregiver dashboards; the Application/Service Layer, which handles authentication, patient data access, report management, educational content, and prediction requests; and the REPLICA Database, which acts as the system's main data repository.

The Prediction Engine, based on the LMM and Gradient Boosting models, is shown as a supporting subsystem rather than as a fourth layer because it is invoked by the Application/Service Layer only when prediction functionality is required. This also separates model inference from the main web application responsibilities.

Outside the runtime web system, ADNI data are used in an offline model-development pipeline for data preparation, model training, and evaluation. The resulting trained model is then deployed to the Prediction Engine for use during prediction, following the general separation between model development and prediction serving described in ML system architectures [25].

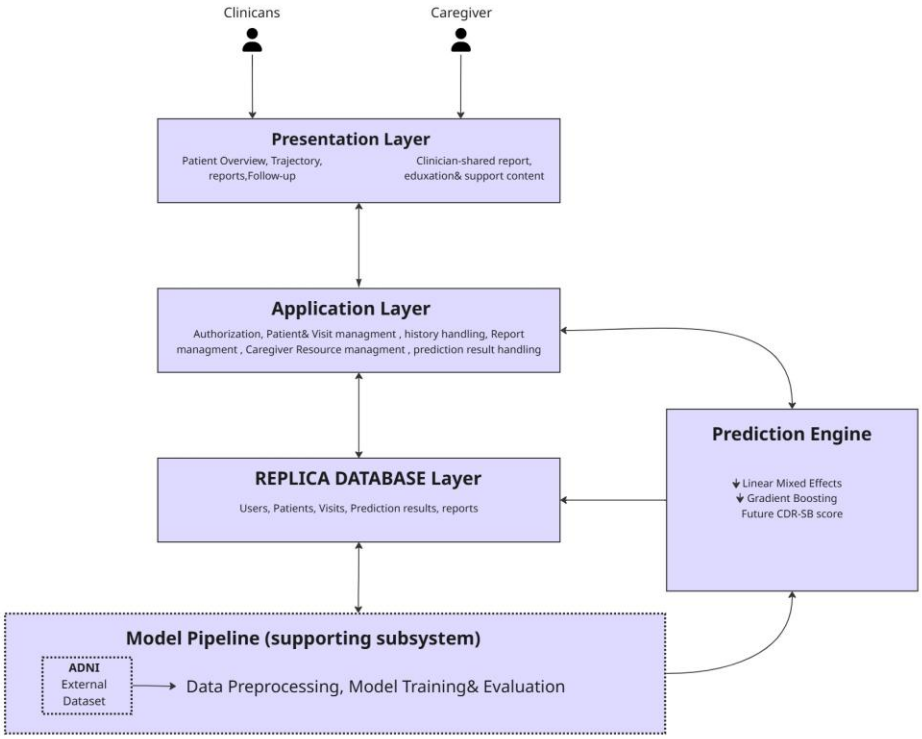


Figure 1: System Architecture

4.4 Use Case Diagram

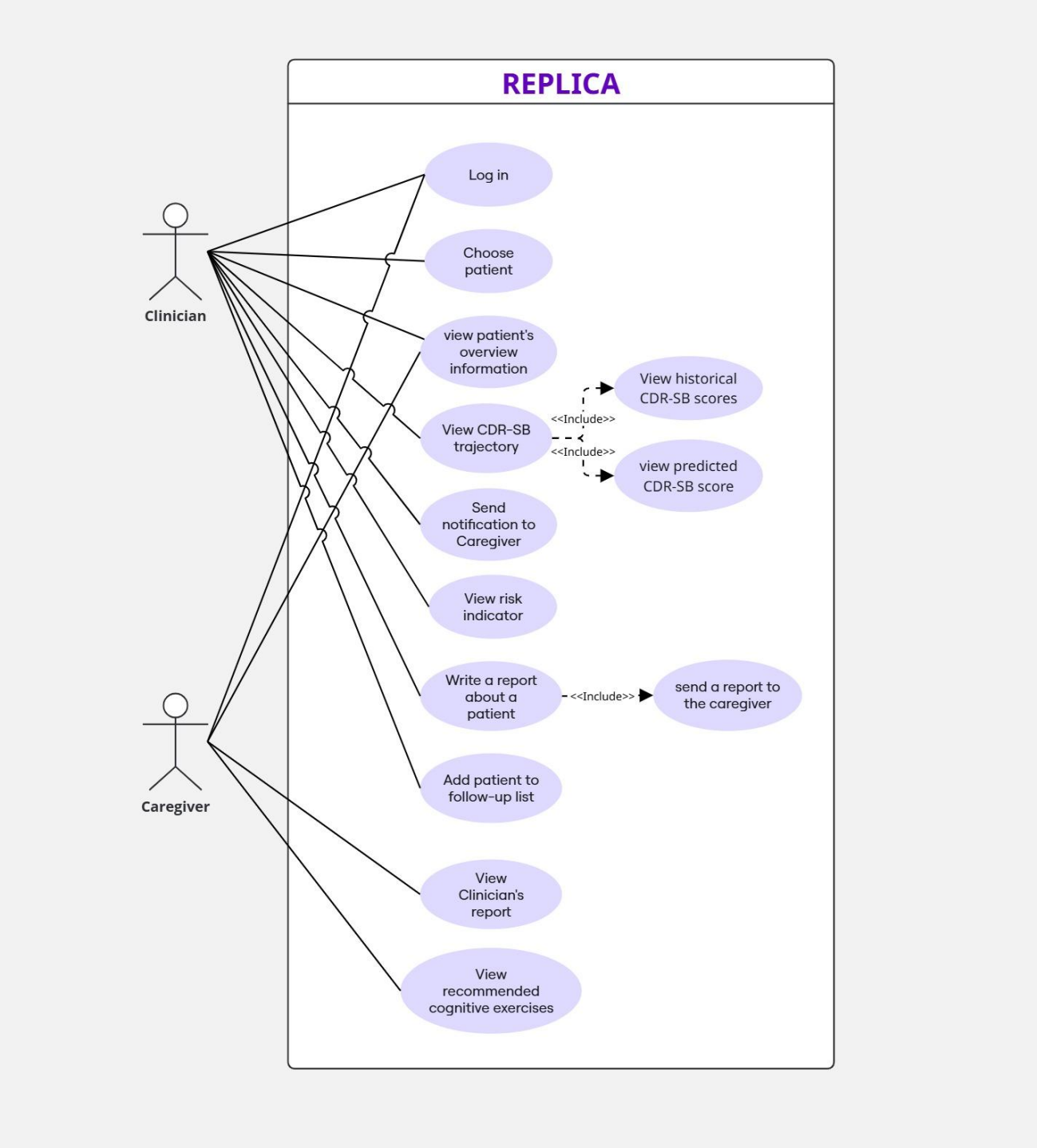


Figure 2: Use Case Diagram

5 Product Backlog

This section contains the prioritized work required to develop REPLICA, including technical work, knowledge acquisition, system features, and non-functional requirements. It covers the data preparation and prediction pipeline, requirements that need further validation, the main functionalities for neurologists and caregivers, and the quality attributes of the system. Each PBI includes its user story, estimated size, type, and acceptance criteria to clarify the expected outcome.

5.1 Product Backlog

ID	PBI (User Story)	Size	Type	Acceptance Criteria
01	When ADNI longitudinal data are prepared for REPLICA, the data-processing pipeline cleans, merges, and organizes the required variables and patient visits for model development.	8	Technical Work	• If ADNI files are processed, then the required patient variables and CDR-SB values are available in the final dataset. • If a patient has multiple visits, then all visits are linked to the correct patient and arranged in chronological order. • If information comes from more than one ADNI file, then the records are matched to the correct patient and visit. • If there are duplicate, missing, or invalid values, then they are handled using the defined preprocessing rules. • If the dataset is split for model training and testing, then the same patient does not appear in more than one split. • If preprocessing is completed, then the final dataset is ready to be used by the prediction model.
02	As a team, I want to determine the minimum visits needed for a reliable prediction and the future horizons to predict, using ADNI cohort analysis	3	Knowledge Acquisition	• If the analysis is completed, then a minimum-visit threshold is documented with supporting justification.

	and expert's knowledge, so that model design is evidence-based, not assumed.			• If the analysis is completed, then a defined set of prediction horizons is documented.
03	When the initial prediction pipeline is developed, REPLICA trains and evaluates a baseline model to establish a reference performance for future model development.	5	Technical Work	• If the baseline model is trained, then it uses the prepared longitudinal dataset. • If a prediction is generated, then the model produces a future CDR-SB prediction for the selected prediction horizon. • If the model is evaluated, then its performance is measured using the defined regression metrics, including MAE and RMSE. • If evaluation is completed, then the baseline results are saved so they can later be compared with the personalized longitudinal model.
04	When longitudinal patient data are available, REPLICA develops and integrates the personalized longitudinal model used to predict future CDR-SB trajectories.	8	Technical Work	• If the longitudinal model is trained, then it uses each patient's previous CDR-SB measurements and their actual visit times. • If patient information is used in the model, then the prediction is generated using the data available for that specific patient. • If a prediction is generated, then the model produces future CDR-SB estimates for the prediction horizons. • If a patient has multiple eligible visits, then the model uses the patient's longitudinal history to estimate their individual progression pattern. • If the model is evaluated, then its performance is measured using the defined regression metrics on patients not used for training. • If evaluation is completed, then the personalized model is compared with the baseline model. • If the model is successfully integrated, then its predicted CDR-SB trajectory can be provided to the clinician dashboard.

05	As a user, I want to log in to REPLICA so that I can securely access the functions available to my role.	3	Feature	• If I enter valid credentials, then I am granted access to the system. • If I enter invalid credentials, then access is rejected and an error message is shown. • If login succeeds, then I am directed to the interface matching my role (neurologist or caregiver).
06	As a neurologist, I want to view, search, and select patients from my patient list so that I can quickly access the patient I need to review.	5	Feature	• If I open my patient list, then all patients assigned to me are displayed. • If I enter a search term, then the list filters to matching patients only. • If I select a patient, then their correct patient information is displayed.
07	As a neurologist, I want to view a patient overview including relevant profile information so that I can understand the patient's current and previous status in one place.	5	Feature	• If I open a patient's overview, then their relevant profile information is displayed. • If the patient has recorded demographic information, then it is shown correctly in the overview. • If I select a different patient, then the overview updates to display only that patient's information. • If any information is unavailable, then the system clearly indicates that it is not available instead of displaying incorrect or empty data.
08	As a neurologist, I want to view the patient's predicted future CDR-SB trajectory alongside historical measurements so that I can review the patient's expected cognitive progression over time.	8	Feature	• If the trajectory is displayed, then historical and predicted CDR-SB values are visually distinguishable. • If I view a patient's trajectory, then the predictions shown correspond to that specific patient.
09	When a new eligible patient visit is added, REPLICA recalculates the patient's future	8	Feature	• If a new visit is added, then it becomes part of the patient's longitudinal history. • If the longitudinal history is updated, then recalculation uses the updated data.

	CDR-SB trajectory using the updated longitudinal history.			• If recalculation completes, then the revised predicted trajectory is displayed to the neurologist.
10	As a neurologist, I want to view the patient's risk indicator based on the predicted CDR-SB trajectory so that I can consider the risk during patient follow-up.	8	Feature	• If I view a patient's record, then a patient-specific risk indicator is displayed. • If the risk is displayed, then it corresponds to the currently selected patient.
11	As a neurologist, I want to manage a follow-up list so that I can easily access patients whom I believe require closer attention.	3	Feature	• If I choose to add a patient to the follow-up list, then the patient is added manually by my action, not automatically by the system. • If I open the follow-up list, then all patients I have added are displayed. • If I remove a patient from the list, then they no longer appear in it.
12	As a neurologist, I want to write and send a patient report with the associated caregiver so that I can communicate relevant patient information.	5	Feature	• If I write and save a report for a patient, then it is stored under that patient's record. • If I share a report with the caregiver, then the caregiver gains access to that specific report. • If a report has not been shared, then the caregiver cannot access it.
13	As a neurologist, I want to send a notification to the associated caregiver so that I can communicate important follow-up information.	3	Feature	• If I create a notification for a patient, then it is sent to that patient's associated caregiver. • If the notification is sent, then it reaches only the correct associated caregiver. • If the caregiver opens the app, then they can view the notification.
14	When the correction layer is implemented, REPLICA applies the Gradient Boosting correction component to improve the longitudinal prediction pipeline and	8	Technical Work	• If the correction layer receives the required inputs, then it produces a corrected prediction. • If evaluation is performed, then performance before and after correction is compared using the same evaluation procedure. • If the correction layer is integrated, then the

	evaluates its effect on prediction performance.			final pipeline can provide predictions to REPLICA. • If the final prediction pipeline is selected, then its output can be provided to the clinician dashboard.
15	As a caregiver, I want to view the patient's profile information made available to me so that I can access relevant information about the patient.	3	Feature	• If I view patient information, then only the correct associated patient's data is shown.
16	As a caregiver, I want to view reports and notification shared by the neurologist so that I can stay informed about important patient information and follow-up.	3	Feature	• If a report or notification has been shared with me, then it appears in my view.
17	As a caregiver, I want to record notes about the patient so that relevant day-to-day observations can be available to me when needed.	3	Feature	• If I write a note, then it is stored under the associated patient's record. • If a note is stored, then it includes the date/time.
18	As a caregiver, I want to access recommended cognitive activities and educational resources so that I can support the patient and learn about MCI and Alzheimer's disease.	3	Feature	• If I open the activities section, then available cognitive activities and educational resources are displayed. • If I select a resource, then it opens and is accessible to me. • If an activity is provided, then it includes sufficient instructions for me to understand and complete it.

Table 6: Product Backlog

5.2 Non-functional requirements

ID	Type	Category	Non-Functional Requirement User Story
01	Feature	Performance	As a clinician, I want patient profiles, historical predicted CDR-SB progression, and risk indicators to be displayed within 5 seconds of selecting a patient, so that I can review the patient's condition without unnecessary delay.
02	Feature	Usability	As a user, I want the system to present information through a clear and consistent interface, so that I can easily understand and access the information relevant to my role without requiring advanced technical or machine-learning knowledge.
03	Feature	Security	As an authorized user, I want patient information and system functions to be protected through secure authentication and role-based access control, so that clinicians and caregivers can access only the information and functions permitted for their roles.
04	Feature	Availability	As a user, I want the system to remain available during its intended demonstration, testing, and usage periods, so that I can access the system and its permitted functions whenever it is being used.
05	Feature	Reliability	As a clinician, I want patient visit records, CDR-SB measurements, predicted progression, risk information, reports, notifications, and follow-up information to be stored and retrieved accurately without unintended data loss, so that I can rely on the system when monitoring a patient over time.

Table 7: Non-functional requirements

Jira:

https://2026-1st-gp16.atlassian.net/jira/software/projects/GP/boards/1/backlog?atlOrigin=eyJpIjoiYThiYTVlYjA5NWFmNDBkYzlhMTFhODI5MmRiMjAxZjEiLCJwIjoiaiJ9

Github:

https://github.com/Danah911/Replica

6 References

[1] National Institute on Aging, "Alzheimer's Disease Fact Sheet," National Institute on Aging, [Online]. Available: https://www.nia.nih.gov/health/alzheimers-and-dementia/alzheimers-disease-fact-sheet. [Accessed 26 September 2026].

[2] World Health Organization, "Dementia," 3 July 2026. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/dementia. [Accessed 26 9 2026].

[3] National Institute on Aging, "What Is Mild Cognitive Impairment?," [Online]. Available: https://www.nia.nih.gov/health/memory-loss-and-forgetfulness/what-mild-cognitive-impairment.

[4] X. Wang, Y. T., W. Ye and J. Zhang, "Uncovering heterogeneous cognitive trajectories in mild cognitive impairment: a data-driven approach," *Alzheimer's Research & Therapy,* vol. 15, no. 1, 2023.

[5] H. Ding, B. Wang, A. P. Hamel, M. Melkonyan, T. F. A. Ang, R. Au and H. Lin, "Prediction of progression from mild cognitive impairment to Alzheimer's disease with longitudinal and multimodal data," *Frontiers in Dementia,* vol. 2, 2023.

[6] Alzheimer's Disease Neuroimaging Initiative, "ADNI Data," [Online]. Available: https://adni.loni.usc.edu/data-samples/adni-data/.

[7] N. M. Laird and J. H. Ware, "Random-effects models for longitudinal data," *Biometrics,* vol. 38, no. 4, pp. 963-974, 1982.

[8] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16)*, New York, NY, USA, 2016.

[9] S. Jamalian, M. Dolton, P. Chanu, V. Ramakrishnan, Y. Franco, K. Wildsmith, P. Manser, E. Teng, J. Y. Jin, A. Quartino and J. C. Hsu, "Modeling Alzheimer's disease progression utilizing clinical trial and ADNI data to predict longitudinal trajectory of CDR-SB," *CPT: Pharmacometrics & Systems Pharmacology,* vol. 12, no. 7, pp. 1029-1042, 2023.

[10] J. C.Morris, "The Clinical Dementia Rating (CDR): Current version and scoring rules," *Neurology,* vol. 43, no. 11, pp. 2412-2414, 1993.

[11] A. M. Wessels, S. A. Dowsett and J. R. Sims, "Detecting Treatment Group Differences in Alzheimer's Disease Clinical Trials: A Comparison of Alzheimer's Disease Assessment Scale - Cognitive Subscale (ADAS-Cog) and the Clinical Dementia Rating - Sum of Boxes (CDR-SB)," *The Journal of Prevention of Alzheimer's Disease,* vol. 5, no. 1, pp. 15-20, 2018.

[12] M. M. Williams, M. Storandt, C. M. Roe and M. John C., "Progression of Alzheimer's disease as measured by Clinical Dementia Rating Sum of Boxes scores," *Alzheimer's & Dementia,* vol. 9, no. 1 Suppl., pp. S39-S44, 2013.

[13] National Institute on Aging, "Alzheimer's Disease Genetics Fact Sheet," 1 March 2023. [Online]. Available: https://www.nia.nih.gov/health/alzheimers-disease-genetics-fact-sheet.

[14] W. M. van Oostveen and E. C. M. de Lange, "Imaging Techniques in Alzheimer's Disease: A Review of Applications in Early Diagnosis and Longitudinal Monitoring," *International Journal of Molecular Sciences,* vol. 22, no. 4, 2021.

[15] Agency for Healthcare Research and Quality, "Clinical Decision Support (CDS)," [Online]. Available: https://www.ahrq.gov/topics/clinical-decision-support-cds.html.

[16] D. Zhang and D. Shen, "Predicting future clinical changes of MCI patients using longitudinal and multimodal biomarkers," *PLoS ONE,* vol. 7, no. 3, 2012.

[17] C. Platero and M. C. Tobar, "Predicting Alzheimer's conversion in mild cognitive impairment patients using longitudinal neuroimaging and clinical markers," *Brain Imaging and Behavior,* vol. 15, no. 4, pp. 1728-1738, 2021.

[18] R. V. Marinescu, N. P. Oxtoby, N. P. Oxtoby, A. L. Young, E. E. Bron, A. W. Toga, M. W. Weiner, F. Barkhof, N. C. Fox, P. Golland, S. Klein and D. C. Alexander, "TADPOLE Challenge: Accurate Alzheimer's Disease Prediction Through Crowdsourced Forecasting of Future Data," in *Predictive Intelligence in Medicine*, 2019.

[19] O. R. Olaniran, S. F. Olaniran, J. Allohibi, A. A. Alharbi and N. M. Alharbi, "Mixed effect gradient boosting for high-dimensional longitudinal data," *Scientific Reports,* vol. 15, no. 1, 2025.

[20] C. Di Lorito, A. Bosco, H. Rai, M. Craven, D. McNally, C. Todd, V. Booth, A. Cowley, L. Howe and R. H. Harwood, "A systematic literature review and meta-analysis on digital health interventions for people living with dementia and Mild Cognitive Impairment," *International Journal of Geriatric Psychiatry,* vol. 37, no. 6, pp. 1-25, 2022.

[21] B. Soykan, G. Hancerliogullari Koksalmis, H.-H. Huang and L. J. Brattain, "CognitiveTwin: Robust Multi-Modal Digital Twins for Predicting Cognitive Decline in Alzheimer's Disease," arXiv, 24 April 2026. [Online]. Available: https://arxiv.org/abs/2604.22428.

[22] C. Chu, Y. Wang, A. L. H. Huynh, K. W. Ng, S. Liu, G. Ji, J. Doecke, J. Fripp, C. L. Masters, B. Goudey, L. Jin and Y. Pan, "Development and validation of a model to predict the progression of Alzheimer's disease," *Age and Ageing,* vol. 54, no. 7, 2025.

[23] M. Bruun, K. S. Frederiksen, H. F. M. Rhodius-Meester, M. Baroni, L. Gjerum, J. Koikkalainen, T. Urhemaa, A. Tolonen, M. van Gils, D. Rueckert, N. Dyremose, B. B. Andersen, A. W. Lemstra and Hallikainen, "Impact of a clinical decision support tool on prediction of progression in early-stage dementia: a prospective validation study," *Alzheimer's Research & Therapy,* vol. 11, no. 1, 2019.

[24] Microsoft, "N-tier architecture style," Microsoft Learn – Azure Architecture Center, [Online]. Available: https://learn.microsoft.com/en-us/azure/architecture/guide/architecture-styles/n-tier.

[25] Google Cloud, "MLOps: Continuous delivery and automation pipelines in machine learning," Google Cloud Architecture Center, [Online]. Available: https://docs.cloud.google.com/architecture/mlops-continuous-delivery-and-automation-pipelines-in-machine-learning.

7 Appendix

Interview scripts:

-Doctors' Script:
Thank you for taking the time to speak with me. I'm [Name], an IT student at King Saud University working on our graduation project, REPLICA, a system that tracks a patient's cognitive status from Mild Cognitive Impairment (MCI) to Alzheimer's Disease and predicts their future progression.

My goal today is to understand how you work with a patient's historical data, and what kind of information display would genuinely help you in practice, so we can design the dashboard around real clinical needs.

This should take about 5-10 minutes. There are no right or wrong answers.

1- Walk me through what a normal follow-up appointment with an MCI patient looks like for you today.

2- When you view a patient's cognitive history, what past information matters most to you (past MMSE scores, CDR, MRI findings, genetic status)?

3- When you see a patient with MCI today, what specifically makes it hard to judge how quickly their condition will change?

4- In your experience, after how many visits do you feel confident forming a real judgment about a patient's trajectory?

5- If I showed you a predicted CDR-SB (Clinical Dementia Rating Sum of Boxes) trajectory for a patient, what would need to be true about it (accuracy level, key factors driving the prediction) for you to actually trust it enough to act on it?

6- Would a single predicted number (e.g., "CDR-SB 12 in 12 months") or a risk category (Low/Moderate/High) be more actionable for you?

7- Thinking about your actual workflow during a follow-up visit, at what specific point would checking a tool like this fit naturally, and what would make it feel like a disruption instead?

That's all my questions, thank you again for your time. Your feedback will genuinely shape how we design this. If anything else comes to mind after we finish, feel free to reach out.

-Caregivers' Script:
Thank you for taking the time to speak with me. I'm [Name], an IT student at King Saud University working on our graduation project, REPLICA, a system that tracks a patient's cognitive status from Mild Cognitive Impairment (MCI) to Alzheimer's Disease and predicts their future progression.

My goal today is to understand your experience caring for someone with memory issues, and what kind of information or tools would genuinely help you, so we can design the caregiver dashboard around real needs.

This should take about 5-10 minutes. There are no right or wrong answers.

1- How do you currently keep track of your patient's memory or condition day-to-day? (notes, memory, an app, nothing formal?)

2- How often do you interact with the patient's doctor, and what information do you usually share with them?

3- Do you currently do any specific activities with them to help their memory or mind (games, puzzles, conversation exercises)? What's worked or not worked?

4- If the app suggested a specific mind activity for your patient today, what would make you actually trust and try it? (e.g., explained why it's suggested, simple to do at home, doctor-approved)

5- Is there anything you'd want from an app like this that we haven't asked about?

That's all my questions, thank you again for your time. Your feedback will genuinely shape how we design this. If anything else comes to mind after we finish, feel free to reach out.

Interview Replies:

Interview 1:

Interviewer: Sondos

Interviewee: Dr.Abdullah

1- Walk me through what a normal follow-up appointment with an MCI patient looks like for you today.

Usually 15-20 minutes, sometimes less if the clinic is busy. I start by asking the family or the patient how things have been since last visit, any new falls, confusion, getting lost, changes in mood. Then I do a quick cognitive screen, MMSE or MoCA depending on what I have time for, review any labs or imaging if it's due, and adjust medication if needed. Honestly a lot of the visit is just listening to the caregiver, because they notice things I can't catch in one appointment.

2- When you view a patient's cognitive history, what past information matters most to you?

The trend in CDR or MoCA scores matters more than any single number, is it stable, dropping slowly, or dropping fast. MMSE if I have it. MRI is important early on to rule out other causes, but I don't repeat it every visit unless something changes. Genetic status, honestly, I rarely have that available here, it's not routine practice for most of our patients.

3- When you see a patient with MCI today, what specifically makes it hard to judge how quickly their condition will change?

The biggest problem is that MCI is not one thing, some patients stay stable for years and some convert to Alzheimer's within twelve months, and on the first visit they can look identical. Comorbidities like diabetes, depression, poor sleep all affect cognition and confuse the picture. And a lot of my patients don't come regularly, they disappear for a year then come back, so I lose the continuity that would help me judge the slope.

4- After how many visits do you feel confident forming a real judgment about a patient's trajectory?

Realistically three to four visits, so over a year to a year and a half, especially if I have consistent cognitive testing each time. One visit tells me almost nothing about trajectory, it just tells me where they are today.

5- What would need to be true about a predicted CDR-SB trajectory for you to trust it enough to act on it?

I'd need to know it's validated on a population similar to mine, our patients often have different risk factors and education levels. I'd also want to see what's driving the prediction, is it mainly cognitive scores, or is it including MRI, comorbidities, lifestyle. A number with no explanation, I won't act on it, I'll just ignore it honestly.

6- Would a single predicted number or a risk category be more actionable for you?

Risk category, definitely. A number like "12 in 12 months" means very little to me on its own without knowing the confidence around it, but "high risk of rapid decline" tells me to act now, refer sooner, counsel the family sooner. Simpler is more usable in a busy clinic.

7- At what point in the visit would checking a tool like this fit naturally, and what would make it feel like a disruption?

It would fit right after I finish the cognitive screen, while I'm reviewing the chart and deciding on next steps, that's when I'm already thinking about trajectory anyway. It becomes a disruption if it needs a lot of manual data entry during the visit, or if I have to leave my normal system to go check another platform. If it's not integrated into what I already use, I probably won't use it consistently, no matter how good the prediction is.

Interview 2:

Interviewer: Sondos

Interviewee: Dr.Hesham

1- Walk me through what a normal follow-up appointment with an MCI patient looks like for you today.

Depends on the day, but on average maybe 10 minutes per patient. I check what changed since last time based on the file, ask the companion a few direct questions, and check if anything needs referral. Not much time for deep conversation unfortunately, that's the reality of government clinics.

2- When you view a patient's cognitive history, what past information matters most to you?

For me it's the score trajectory first, second is functional status, can they still manage their medications, drive, handle money. MRI matters once, at baseline, mainly to exclude vascular or structural causes. Genetic testing, we almost never have it, so I don't rely on it in practice.

3- When you see a patient with MCI today, what specifically makes it hard to judge how quickly their condition will change?

Inconsistent follow-up is my biggest issue. Patients miss appointments, come back six months late, and then I'm comparing two points that aren't even reliable because the testing conditions were different, different time of day, patient was tired. Education level here also skews the cognitive tests, so a low score doesn't always mean what it means in a textbook.

4- After how many visits do you feel confident forming a real judgment about a patient's trajectory?

Minimum two to three, but I'm more confident by the fourth. Anything before that, I'm cautious about telling the family anything definitive, because I've been wrong before when I judged too early.

5- What would need to be true about a predicted CDR-SB trajectory for you to trust it enough to act on it?

I'd want to see the error margin clearly, not just a clean number, because clean numbers give false confidence. I'd also want to know what data it's built on, if it's mostly Western I'd be cautious applying it directly here. Show me the reasoning, not just the output.

6- Would a single predicted number or a risk category be more actionable for you?

Category. A number invites arguments with families "why 12 and not 10", but a category like Low, Moderate, High is easier to communicate and easier to act on quickly.

7- At what point in the visit would checking a tool like this fit naturally, and what would make it feel like a disruption?

Before I walk into the room, ideally, while I'm reviewing the file. If I have to open it mid-consultation it slows me down and the patient notices I'm distracted. It has to be fast, one glance, or I won't use it during a busy clinic day.

Interview 3:

Interviewer: Sondos

Interviewee: Noura a Caregiver for her mother.

1- How do you currently keep track of your patient's memory or condition day-to-day?

Honestly, nothing formal. My mother has mild memory issues so I just keep notes in my phone, sometimes on WhatsApp to myself so I don't lose them. Like if she forgot the name of one of my kids, or asked the same question three times in one hour, I write the date and what happened. No app or anything organized, just whenever I remember to write it.

2- How often do you interact with the patient's doctor, and what information do you usually share with them?

We go every couple of months to the doctor, unless something changes suddenly. Before the appointment I try to go back through my phone notes and remember the worst moments, did she get lost, did she leave the stove on, things like that. Honestly half the time I forget to mention something important because I didn't write it down properly, and I only remember in the car on the way home.

3- Do you currently do any specific activities with them to help their memory or mind?

We try. My sister downloaded some brain games for her but she gets frustrated with the phone screen, it's not really for her generation. What actually works better is just sitting with her and going through old photos, or she likes it when I ask her to tell me recipes step by step. Puzzles she liked at first then lost interest after a week.

4- If the app suggested a specific mind activity for your patient today, what would make you actually trust and try it?

I'd need to know why it's picked this exact activity for her, not just a random game. And it has to be something I can do without a lot of setup, we don't have time for complicated things. If it said something like "her doctor's plan supports this" or it was linked somehow to what the doctor already said, I'd trust it a lot more than just an app deciding on its own.

5- Is there anything you'd want from an app like this that we haven't asked about?

Yes, I really want something that shows me if she's getting worse or staying the same over time, like a simple trend, not just numbers I don't understand. And maybe something I can literally show the doctor on my phone instead of trying to explain from memory. Also, if it could remind me of things, because honestly I forget too, I'm exhausted most days.